COL 774 – Machine Learning – Assignment 1 Report

1 Created Features and Modeling Strategy

Part (a): Baseline linear regression. Ordinary least squares is fit directly on the provided feature columns via the closed-form normal equation on the bias-augmented design matrix X: $\mathbf{w} = (X^\top X)^{-1} X^\top \mathbf{y}$.

Part (b): Ridge-regularized regression. Same closed-form approach, with an L_2 penalty on all weights except the bias: $\mathbf{w} = (X^\top X + \lambda R)^{-1} X^\top \mathbf{y}$, where R is the identity with R_{00}=0. λ is selected by 5-fold cross-validation over the candidate values, choosing the one with lowest mean held-out NMSE, then refitting on the full training set.

Part (c): Heart-rate prediction. Every example has ten seconds of BVP (640 samples), three-axis accelerometer (320 samples/axis) and EDA (40 samples). Instead of feeding these 1640 raw numbers directly into a regressor, 81 hand-built physiological summary features are computed per window, and the final predictor stays linear in this feature vector $\mathbf{z}$: $\hat{y} = w_0 + \mathbf{w}^\top \mathbf{z}$. The 81 base features break down as:

- **BVP shape (9):** mean, RMS, MAD, IQR, 90–10 spread, peak-to-peak range, crest factor, skewness, kurtosis.
- **Derivatives VPG/APG (14):** first derivative (VPG) summarised by mean-abs, variance, skew, kurtosis, max, min, zero-crossings, line length; second derivative (APG) by variance, mean-abs, skew, kurtosis, zero-crossings, turning-point ratio.
- **Cardiac autocorrelation (10):** BVP autocorrelation at lags 32, 38, 48, 64, 77 (64 Hz), bracketing 50–120 bpm, plus a high/low-lag ratio, entropy, TKEO mean, and Hjorth mobility/complexity.
- **Motion (33):** SMA, MAD, RMS, crest factor, CV, vibration intensity of accelerometer magnitude; jerk/snap; per-axis means, roll/pitch, cross-axis correlation, dominant-axis ratio, postural transitions; magnitude skew, kurtosis, IQR, percentiles, zero-crossings, turning ratio, autocorrelation, Hjorth stats, entropy.
- **EDA (15):** level stats (mean/std/min/max/IQR/RMS), skew, kurtosis, slope proxy, line length, derivative zero-crossings, peak-time, tonic/phasic split with phasic energy/max/std.

All 81 features (incl. self-products) are expanded into every pairwise product (3402 columns), standardised, and narrowed to 500 columns by a Lasso (α=0.001) fit on a 100k-row training subsample, used purely as a feature selector. The final predictor is an `SGDRegressor` with epsilon-insensitive loss (ϵ=0, directly minimising MAE) and L_2 penalty, with α chosen by 3-fold CV on negative MAE.

Part (d): CGM glucose estimation. Each example is a five-minute multimodal window from the E4 (BVP/HR/EDA/temp/acc) and Zephyr (ECG/breathing/acc), summarised into 129 features:

- **Cardiovascular timing:** peaks detected on chest ECG and wrist BVP give mean RR, SDNN, RMSSD, pNN50/NN50, Poincaré SD1/SD2, plus Welch-PSD VLF/LF/HF band powers and LF/HF ratio.
- **HR, respiration, temperature:** level, spread, slope and 6-block trend on E4 HR, Zephyr breathing (+peak-counted rate), and skin temperature.
- **EDA:** level statistics plus a Butterworth tonic/phasic split with SCR count and amplitude.
- **Motion:** E4 and Zephyr accelerometer-magnitude statistics (mean/std/range/IQR/skew/kurt, TKEO, zero-crossings, block trend).
- **Cross-modal interactions:** 7 hand-picked products, e.g. RMSSD×SCR-count, HRV×breathing rate.

Features are median-imputed, standardised, and univariate-selected to 80 (mutual information for d1; F-test for d2/d3), degree-2 polynomial expanded (3320 columns), then re-selected to under 500 columns. The final model is a linear `SGDRegressor`: L_2 penalty for d1, ElasticNet with a 5 mg/dL epsilon-insensitive tolerance for d2/d3 (sensor noise floor), with hyperparameters chosen by 3-fold CV on negative MAE, fit on the training folder only.

2 Normalized Acceleration

For accelerometer samples $(a_x(t), a_y(t), a_z(t))$, the squared acceleration magnitude is

$$a_{\text{sq}}(t) = a_x(t)^2 + a_y(t)^2 + a_z(t)^2 \tag{1}$$

and the normalized squared acceleration required by the assignment is

$$\widetilde{a}_{\text{sq}}(t) = \frac{a_{\text{sq}}(t)}{\frac{1}{T}\sum_{\tau=1}^{T} a_{\text{sq}}(\tau)} \tag{2}$$

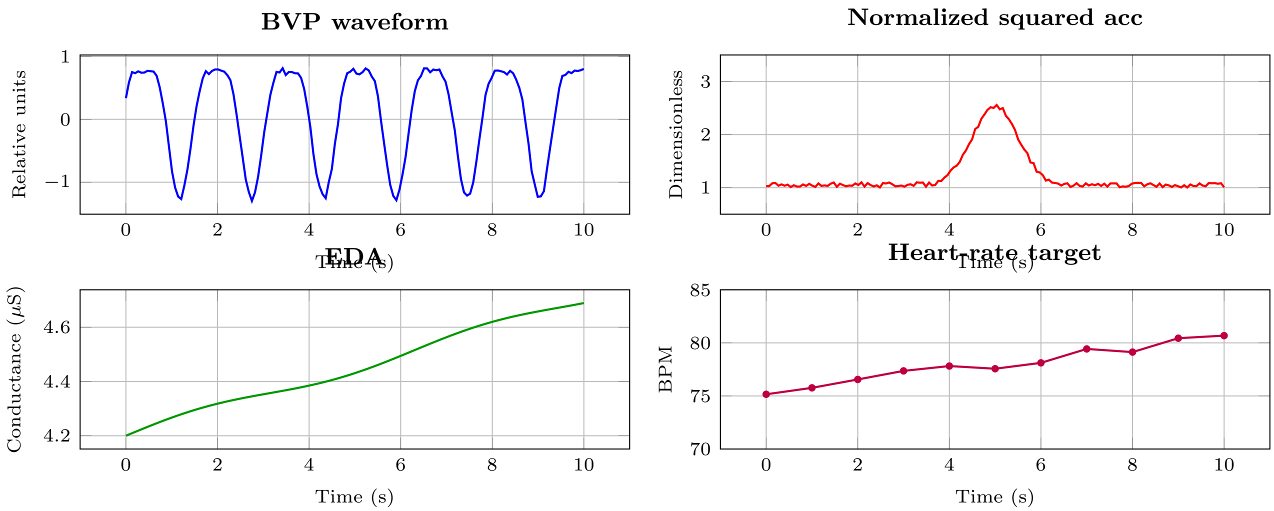

Figure 1: Representative ten-second physiological window for Part (a).

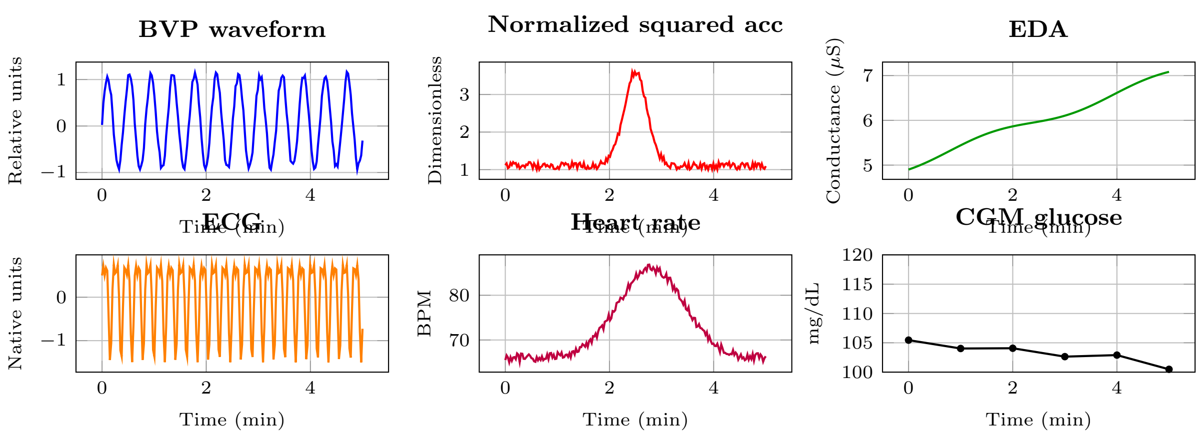

Figure 2: Representative five-minute physiological window for Part (d).

3 Top Five Features

Feature importance is ranked by the (signed) regression coefficient on standardised features, as required.

Part (c): Heart-rate prediction

Feature	Modality	coef	Interpretation
`acc_iqr`	ACC	+7.4617	Wider spread in accelerometer magnitude (more variable motion) predicts *higher* heart rate – motion intensity is a strong confounder/driver of measured BPM.
`acc_entropy`	ACC	+7.4075	More irregular, less periodic motion predicts *higher* heart rate, consistent with physical activity elevating HR.
`vpg_mean_abs*eda_mean`	BVP-deriv/EDA	+7.3594	Pulse-wave derivative magnitude combined with skin-conductance level predicts *higher* heart rate – a joint cardiovascular/sympathetic-arousal signal.
`acc_mean_y*acc_iqr`	ACC	+6.7499	Average lateral (y-axis) acceleration combined with motion variability predicts *higher* heart rate.
`acc_mean_y*acc_mean_z`	ACC	−6.4110	The interaction between mean y- and z-axis acceleration (arm orientation) predicts *lower* heart rate.

Table 1: Top five Part (c) features, n=53,000. Motion terms dominate over the cardiac-autocorrelation features the design was originally centred on.

Protocol	Feature	Modality	coef	Interpretation
D1 (n=3713)	`ecg_nn50`	ECG	−4.4047	More large beat-to-beat interval jumps predicts *lower* glucose – a direct HRV/autonomic marker.
	`ecg_mean_rr*zacc_zcross`	ECG/Chest-ACC	−4.2655	Longer heartbeat spacing combined with chest-motion frequency predicts *lower* glucose.
	`zacc_zcross^2`	Chest-ACC	+4.0639	Nonlinear term on chest-motion frequency predicts *higher* glucose.
	`breath_abs_diff_mean*zacc_zcross`	Breath/Chest-ACC	−3.4999	Breathing-signal roughness combined with chest motion predicts *lower* glucose.
	`bvp_mean_rr`	BVP	+3.2427	Longer pulse-to-pulse spacing predicts *higher* glucose – a direct heart-rate proxy.
D2 (n=3317)	`ecg_pnn50`	ECG	−3.5698	Same autonomic marker as D1's `ecg_nn50` – higher fraction of large beat jumps predicts *lower* glucose.
	`breath_max*eda_x_temp`	Breath/EDA/Temp	−3.0086	Peak breathing amplitude combined with EDA–temperature interaction predicts *lower* glucose.
	`eda_min*bvp_std`	EDA/BVP	+2.6918	Minimum skin conductance combined with pulse-wave variability predicts *higher* glucose.
	`ecg_mean_rr^2`	ECG	−2.4538	Nonlinear term on average heartbeat spacing predicts *lower* glucose.
	`ecg_range*zacc_zcross`	ECG/Chest-ACC	+2.4140	ECG range combined with chest-motion frequency predicts *higher* glucose.
D3 (n=3757)	`ecg_pnn50`	ECG	−3.5711	Same top marker as D2, now the single strongest driver under cross-subject generalisation.
	`ecg_pnn50*bvp_std`	ECG/BVP	−2.5292	The same HRV marker interacting with wrist pulse-wave variability.
	`breath_abs_diff_mean*zacc_mean`	Breath/Chest-ACC	−2.4870	Breathing-signal roughness combined with average chest-motion level.
	`breath_abs_diff_mean`	Breath	−2.4804	Raw respiratory variability, with no interaction term – the model leans on this directly once subjects are unseen.
	`temp_min^2`	Temperature	−2.3911	Nonlinear term on minimum skin temperature.

Table 2: Top five Part (d) features per protocol. ECG-derived HRV (`ecg_nn50`/`pnn50`) and chest-accelerometer zero-crossings recur across all three protocols, but D2/D3 lean more heavily on breathing and temperature terms than D1 – plausibly because the random within-subject split (D1) lets the model partly memorise subject-specific cardiac baselines, while the harder temporal (D2) and cross-subject (D3) splits push it toward more slowly-varying, subject-general signals.

Part (d): CGM glucose estimation

4 Median Baseline Comparison

For an evaluation set with mean target $\bar{y}$, the assignment defines

$$\mathrm{NMAE} = \frac{\sum_i |y_i - \hat{y}_i|}{\sum_i |y_i - \bar{y}|} \quad \text{and} \quad \mathrm{NMSE} = \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{3}$$

The required baseline predicts the median of the training targets for every validation example.

	Median baseline		Final model	
Part / Protocol	NMAE	NMSE	NMAE	NMSE
(c) Heart rate (n=53,000)	0.9789	1.0162	**0.5730**	**0.4865**
(d) D1 – random within-subject (n=3713)	0.9692	1.0564	**0.8726**	**0.9078**
(d) D2 – temporal within-subject (n=3317)	0.9951	1.1429	**0.9467**	**1.0795**
(d) D3 – cross-subject (n=3757)	1.0958	1.3660	1.1657	1.4901

Table 3: Part (c) and Part (d) model vs. median-target baseline.

Part (c) is the strongest result: NMAE drops from a 0.979 median baseline to 0.573, with motion terms dominating the top-5 coefficients rather than the cardiac-autocorrelation features the design centred on, suggesting motion is doing much of the predictive work as a heart-rate confounder. Part (d) shows a clear difficulty gradient: D1 comfortably beats its baseline, D2 still beats its baseline but by a smaller margin, and **D3 fails to beat its baseline on both metrics** – plausibly because the 500 selected interaction features overfit subject-specific patterns across only 8 training subjects and don't transfer to the 2 held-out ones (`c1s04`, `c2s03`), leaving little room for 3-fold CV to guard against it. D2/D3 training also triggered `scipy` precision-loss warnings during skew/kurtosis computation on near-constant windows, whose NaNs are absorbed by the median imputer without propagating to the final model. The main open item is whether D3 needs stronger regularisation, fewer selected features, or subject-level normalisation to close the baseline gap.